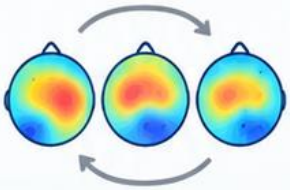
EEG Microstates:
A Compact View of Topographic Dynamics

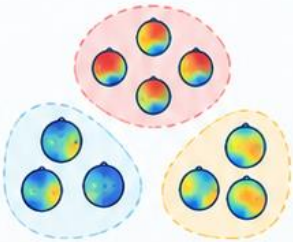
Handout / Poster Companion

1 Recurring topographies



EEG scalp maps change continuously, but some spatial configurations recur. Around moments of high global field power (GFP), these dominant topographies can be especially clear and reproducible.

2 A low-dimensional summary



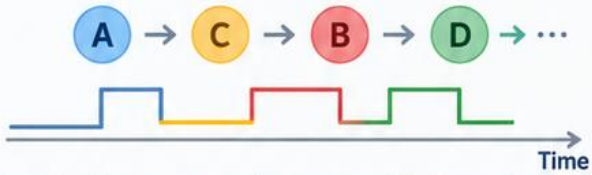
Microstate analysis clusters similar scalp topographies into a small number of representative classes. This provides a low-dimensional description of the continuously evolving global EEG field.

3 Dominant, not exclusive



A microstate class reflects the currently dominant topographic configuration. It does not imply that the whole brain is in a single exclusive physiological state; multiple generators and networks may contribute in parallel.

4 Temporal dynamics become measurable



Discretization turns ongoing topographic fluctuation into a symbolic sequence (A → C → B → D ...), allowing analysis of duration, occurrence, coverage, transitions, syntax, entropy, and higher-order structure.

5 Only part of the EEG state



Topography (spatial configuration)
Amplitude
Spectral content
Phase
Connectivity
Source mixtures

The class label mainly captures spatial configuration. Other relevant features — amplitude, spectral content, phase, connectivity, and source mixtures — are not fully represented by the label alone.

6 A useful coarse-grained representation



Microstates are neither arbitrary labels nor complete physiological states. They are recurring dominant configurations of the global EEG field that provide a compact, physiologically meaningful, coarse-grained representation of brain dynamics.

Take-home message: Microstates capture dominant recurring configurations of the EEG field — a useful coarse-grained representation of brain dynamics, rather than a complete description of the underlying physiological state.

Interactive visualization
https://christoph-berger.github.io/microstates/

Scan to explore